

- A relation in which one quantity increases in the same proportion by decreasing the other quantity and vice versa is called the inverse proportion.
- Time is directly proportional to the work and the number of workers is inversely proportional to time.
- **Speed:** The ratio of distance to time is called speed. It can be calculated by the formula.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

The distance covered per unit time is called speed.

- To understand the relation between distance, speed and time, the formula is:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Conversion of units of speed

- To convert km/h into m/s , multiply with $\frac{1000}{3600}$
- To convert m/s into km/h , divide by $\frac{1000}{3600}$

Think Tank: The Most Wonderful Knowledge.

Ask your friend to take any 3 digit number in the descending order of digits. Reverse it and then subtract it from the original number. Again reverse the result and add. You tell him, "your answer is 1089". He will be surprised!

$$\begin{array}{r} 732 \\ - 237 \\ \hline 495 \\ + 594 \\ \hline 1089 \end{array}$$

$$\begin{array}{r} 321 \\ - 123 \\ \hline 198 \\ + 891 \\ \hline 1089 \end{array}$$

Similarly, try with other such numbers.

Introduction

In our daily life, we use different facilities provided by the state like police for security, government schools for education, government Hospitals for health etc. Government of the state provides such facilities by an amount which is collected from the citizen of the country and this amount is called **tax**.

7.1 Taxes

A fee charged (levied) by the government on a product, income or activity is called tax. If tax is levied directly on personal or corporate income, then it is called **direct tax**. If tax is levied on the price of goods or services, then it is called **indirect tax**.

The purpose of taxation is to finance government expenditure. One of the most important use of taxes is to finance public goods and services.

7.1.1 Property Tax and General Sales Tax (GST).

There are many types of taxes charged by the government, such as wealth tax, income tax, property tax, general sales tax etc.

Here we should study only the following two types of taxes:

- (i) Property Tax
- (ii) General Sales Tax (GST)

(i) Property Tax

Property tax is a levy issued by a government on a person's real or personal property. The property is assessed to give it a value and then that value is taxed.

The amount of tax on a given property may change over time leased on a re-assessment of the property value.

Property tax payments are due annually in most cases.

It is generally paid at a flat rate of 2% but the tax rates vary, depending on the province.

In our province property tax is levied and collected under Sindh Urban Property Tax Act 1958.

Following properties are exempted from tax.

1. Government owned properties.
2. Residential building on land not over 120 square yards.

3. One residential flat of 600 square feet or less on any floor of a building.
4. Libraries, parks, playgrounds, place of worship, orphanages, cemeteries, crematorium etc.
5. Properties owned by widows, orphaned children and disabled individuals within some specified limit.
6. Properties listed as protected heritage under the Sindh Cultural Heritage Preservation Act 1994.

Example 1. Find the property tax on a property of Rs 1,400, 000 at the rate of 3%.

Solution: Worth of the property = Rs 1,400,000

Tax rate = 3%

Property tax = ?

Formula for finding property tax = $\frac{\text{Rate} \times \text{value of property}}{100}$

Property tax = 3% of Rs 1,400,000 = $\frac{3}{100} \times 1,400,000$

$$= \frac{3 \times 1,400,000}{100}$$

$$= \frac{3 \times 1,400,000}{100} = 3 \times 14000 = \text{Rs } 42000$$

Thus the required property tax is Rs 42,000.

Example 2. A lady doctor paid Rs 9500 as a property tax at the rate of 2%. Find the worth of her property.

Solution: Property tax paid = Rs 9500

Tax rate = 2%

Worth of the property = ?

2% of the worth of property = Rs 9500

1% of the worth of property = $\left(\frac{\text{Rs } 9500}{2} \right)$

100% of the worth of property = $\text{Rs } \left(\frac{9500 \times 100}{2} \right) = \text{Rs } \frac{9500 \times 100}{2}$
 $= \text{Rs } 9500 \times 50 = \text{Rs } 475,000$

Thus the required worth of lady doctor's property is Rs 475,000.

EXERCISE 7.1**A. Find property tax on the property at the rate mentioned below:**

1. Property worth Rs 286,400, rate = 5%.
2. Property worth Rs 1,657,900, rate = 2%.
3. Property worth Rs 814,500, rate = 3%.
4. Property worth Rs 234,000, rate = 1.4%.
5. Property worth Rs 2,050,000, rate = 1.6%.

B. Find the worth of the property in the following when:

1. Tax paid on the property is Rs 25500 at the rate of 2%.
2. Tax paid on the property is Rs 35900 at the rate of 4%.
3. Tax paid on the property is Rs 4800 at the rate of 5%.
4. Tax paid on the property is Rs 8500 at the rate of 3.5%.
5. Tax paid on the property is Rs 11,700 at the rate 1.5%.

C. Find rate percentage of property tax when:

1. The property tax paid is Rs 30000 and the worth of the property tax paid is Rs 1,800,000.
2. The property tax paid is Rs 33000 and the worth of the property is Rs 3,300,000.
3. The property tax paid is Rs 12000 and the worth of the property is Rs 600,000.
4. The property tax paid is Rs 8676 and the worth of the property is Rs 433,800.
5. The property tax paid is Rs 12000 and the worth of the property is Rs 600,000.

D. Answer the following:

1. What is tax?
2. What is the purpose of taxation?
3. Explain property tax.
4. Mention any four properties which are exempted from property tax.

(ii) General Sales Tax (GST) or Sales Tax

A sales tax is a tax paid to the Government for the sales of certain goods and services. Usually seller is allowed to collect funds for the tax from consumer at the point of purchase.

Sellers some times itemize the tax separately from the price of the goods or services, or include it in the price (tax-intensive).

Often exemption is provided from general sales tax on certain goods or services.

7.1.2 Calculation of General Sales Tax (GST)

Since the year 2014 the rate of GST in Pakistan is 17%.

The GST is calculated as under:

$$\begin{aligned}\text{Total GST} &= \text{Item Cost} \times \text{Rate of GST} \\ \text{So, Rate of GST} &= \text{Total GST Tax} / \text{Item Cost}\end{aligned}$$

Example: The cost of a pair of shoes is Rs 2500. The seller charges 17% GST on the cost. Find GST levied.

Solution:

$$\begin{aligned}\text{Cost of item} &= \text{Rs } 2,500 \\ \text{Rate of GST} &= 17\% \\ \text{Value of GST} &= \text{Item Cost} \times 0.17 \\ &= \text{Rs } 2,500 \times 0.17 = \text{Rs } 425\end{aligned}$$

So, the GST charged is Rs 425.

(i) To Determine the Total Cost

In order to determine the total cost and tax inclusive of an item, the GST is added to the item cost.

So, $\text{Total Item Cost} = \text{GST} + \text{Item Cost}$

In the above example;

$$\begin{aligned}\text{Item Cost} &= \text{Rs } 2,500 \\ \text{Value of GST} &= \text{Rs } 425 \\ \text{Therefore, the total cost of pair of shoes} &= \text{Rs } 2,500 + \text{Rs } 425 = \text{Rs } 2,925\end{aligned}$$

So, the total cost of shoes is Rs 2,925

$$\text{GST} = \text{Total Cost} - \text{Item Cost}$$

and $\text{Item Cost} = \text{Total Cost} - \text{GST}$

7.1.3 To Determine Item Cost when Total Cost and Rate of GST is known.

We know that GST is at the rate of 17% of the item cost and is included in total cost.

Let item cost = y

Since Rate of GST = 17% = 0.17

So, GST = $0.17 \times y = 0.17y$

As we know that

Item Cost = Total Cost - GST

Therefore $y = \text{Total Cost} - 0.17 \times y$

or Total Cost = $y + 0.17y = (1 + 0.17)y$

or $1.17y = \text{Total Cost}$

or $y = \frac{\text{Total Cost}}{1.17}$

Hence, $\boxed{\text{Item Cost} = \frac{\text{Total Cost}}{1.17}}$, where Rate of GST is 17%

Example 1: A washing machine is sold for Rs 15,912, inclusive of GST. Find the cost of washing machine excluding GST when rate of GST is 17%.

Solution:

We know that

$$\text{Item Cost} = \frac{\text{Total Cost}}{1.17}$$

In this problem:

Cost of washing machine = Item cost = ?

Total cost = Rs 15,912

Rate of GST = 17% = 0.17

So, Cost washing machine = Rs $\left(\frac{15,912}{1.17}\right) = \text{Rs } 13,600$

Thus, the cost of washing machine (exclusive of tax) is Rs 13,600.

Example 2: A company pays Rs 380 as GST on the services of cost Rs 2,180. Find the rate of the GST.

Solution:

We know that: Rate of GST = $\frac{\text{GST}}{\text{Item Cost}}$

Here we have to find Rate of GST = ?

$$\text{(GST) Paid} = \text{Rs } 380$$

$$\text{Item cost} = \text{Service cost} = \text{Rs } 2,180$$

$$\text{So, Rate of GST} = \frac{\text{Rs } 380}{\text{Rs } 2,180} = 0.17 = \frac{17}{100} = 17\%$$

Thus the Company paid 17% GST.

Example 3: A Gas company charges Rs 3,212 as gas charges from consumers. Find the amount of GST at the rate of 17%, paid by the consumers.

Solution:

We know that

$$\text{GST} = \text{Item Cost} \times 17\%$$

$$\text{Here GST} = ?$$

$$\text{Item cost} = \text{Gas charges} = \text{Rs } 3,212$$

$$\text{Rate of GST} = 17\%$$

$$\begin{aligned} \text{So, GST} &= \text{Rs } 3,212 \times 17\% = \text{Rs } 3,212 \times 0.17 \\ &= \text{Rs } 546.04 = \text{Rs } 546 \end{aligned}$$

Hence GST paid by the consumer is Rs 546.

Example 4: Mr. Adam paid Rs 595 as GST at the rate of 17% on the purchase of a shirt. Find the cost of the shirt exclusive of GST.

Solution:

$$\text{We know that: GST} = \text{Item Cost} \times 17\%$$

$$\text{Here Cost of shirt} = \text{Item Cost} = ?$$

$$\text{GST} = \text{Rs } 595$$

$$\text{Rate of GST} = 17\% = 0.17$$

$$\text{So, } 595 = \text{Cost of shirt} \times 0.17$$

$$\text{or Cost of shirt} = \frac{\text{Rs } 595}{0.17} = \text{Rs } 3,500$$

Thus the cost of shirt is Rs 3500.

Example 5: A person purchased some furniture items worth Rs 77,950 including GST. Find the actual cost of the furniture items and amount of GST at the rate of 17% paid by him.

Solution:

Here; Item Cost = Cost of Furniture = ?

Total Cost of Furniture items = Rs 77,950

Rate of GST = 17% = 0.17

$$\begin{aligned}\text{Now, Cost of Furniture items} &= \frac{\text{Total Cost}}{1.17} = \frac{77,950}{1.17} \\ &= \text{Rs } 66,623.93 = \text{Rs } 6,6624 \\ \text{Amount GST} &= 66624 \times 0.17 \\ &= \text{Rs } 11,326.08 = \text{Rs } 11,326\end{aligned}$$

Thus the total cost of furniture items is Rs 66624 and the amount of GST is Rs 11326.

EXERCISE 7.2

1. The cost of a parker pen is Rs 875. Find the GST charged on it at the rate of 17%.
2. The cost of a mobile is Rs 4,850. Find the GST at the rate of 17%.
3. The total cost of cosmetic items is Rs 14976. If the actual cosmetic item cost is Rs 12,800, find the amount of GST.
4. A microwave costs Rs 15200. Rate of GST is 17% charged on it. Find the amount of GST and total cost of the microwave.
5. Abdullah purchased a three piece suit of cost Rs 7,800. Find the total cost and amount of GST at the rate of 17%.
6. A refrigerator is sold inclusive of 17% GST for Rs 35,375. Find the actual cost of the refrigerator.
7. A person purchased a dishwasher for Rs 61,000 inclusive 17% GST. Find the cost of dishwasher excluding GST. Also find the amount of GST.
8. Ms. Alia paid Rs 2,720 as GST at the rate of 17% on the purchase of a decoration piece. Find the cost of decoration piece exclusive of GST.

9. An electric company charges Rs 1547 GST on the consumption of electricity of the cost Rs 9,100. Find the rate of GST.
10. The cost of a shirt excluding tax is Rs 2,560, and the amount of GST on it is Rs 436. Find rate of GST.

7.2 PROFIT AND MARKUP

7.2.1 Explain profit and markup

Profit

Profit is the amount of gain received from any business, after deducting all cost and expenses from the total sale.

Profit is also referred as net profit or net earning.

So, profit is the difference between Selling Price (SP) and Cost Price (CP).

$$\text{i.e. } \boxed{\text{Profit} = \text{Selling Price} - \text{Cost Price}}$$

Markup

In our daily life, we often borrow money from our friends and relatives to buy a thing that we repay them after a certain period. Some banks and retail organization also provide the same services and charge an additional amount called **markup**.

7.2.2 Find the rate of profit and markup per annum

Profit Rate

Profit rate is a profitability ratio that shows the relationship between gross profit and total net sales revenue. It is a common measure to evaluate the operational performance of the business.

In other words, it is the percentage of income that is **profit**.

$$\boxed{\text{Profit Rate} = \frac{\text{Gross Profit}}{\text{Net Sale}} \times 100}$$

Gross profit = Net Sale – Cost of goods

So, Selling Price = Cost Price + Cost Price × Profit Rate
 = Cost Price (1 + Profit Rate)

$$\boxed{\text{SP} = \text{CP} (1 + \text{Profit Rate})}$$

and

$$\boxed{\text{Cost Price} = \frac{\text{Selling Price}}{1 + \text{Profit rate}}}$$

Example. Find the profit rate when gross profit is Rs 400 and net sale is Rs 1600.

Solution: Profit Rate = $\frac{\text{Gross Profit}}{\text{Net Sale}} \times 100$

$$= \frac{400}{1600} \times 100$$

$$= \frac{400 \times 100}{1600} = 25$$

Thus Profit Rate = 25%

Markup of per annum

Usually, we calculate the markup as a percentage of the actual amount paid for the things. This is called the **markup rate** and paid amount itself is called the **principal**.

Example: Find the markup on a principal amount of Rs 50,000 for 2 years at the rate of 12%.

Solution:

$$\text{Markup} = \frac{P \times R \times T}{100} = \frac{50000 \times 12 \times 2}{100} = \frac{50000 \times 12 \times 2}{100} = \text{Rs } 112,000$$

Activity 1. Find the missing quantities by using formula.

S.No.	Net Profit	S.P	C.P	Formula
(i)		Rs 1750	Rs 1600	Profit = S.P – C.P
(ii)	Rs 600		Rs 2000	S.P = C.P + Profit
(iii)	Rs 950	Rs 10,450		S.P – Profit = C.P

Activity 2. Find the missing quantities by formula.

S.P	C.P	Net Profit	Profit Rate	Formula
Rs 1000	Rs 800	Rs 200		Profit Rate = $\frac{\text{Profit}}{\text{S.P}} \times 100$
	Rs 1000	Rs 100	10%	S.P = C.P (1 + Profit rate)
Rs 2000	Rs 1600	Rs 400		Profit Rate = $\frac{\text{Profit}}{\text{C.P}} \times 100$
Rs 1200	Rs 1000		20%	
Rs 2750	Rs 2500	Rs 250		

Activity 3:

Find the missing quantities by using the formula.

S.No.	Markup	Principal	Time Period	Markup rate	Formula used
(i)		Rs 500	2 years	12%	$MP = \frac{P \times R \times T}{100}$
(ii)	Rs 205		1 years	8%	$P = \frac{MP \times 100}{R \times T}$
(iii)	Rs 528	Rs 1,650	10 years		$MPR = \frac{MP \times 100}{P \times T}$
(iv)	Rs 350	Rs 3,500		2.5%	$T = \frac{P \times R}{MP \times 100}$
(v)		Rs 100,000	3 years	1.25%	$MP = \frac{P \times R \times T}{100}$
(vi)	Rs 1,050		5 years	4.5%	$P = \frac{MP \times 100}{R \times T}$

7.2.3 (i) Solve real life problems involving profit

Consider the following examples:

Example 1: A person purchased a certain item from wholesaler for Rs 400. Then he sells it to a consumer for Rs 700. What is his profit? Also find his profit rate.

Solution:

We know that

$$\text{Profit} = \text{Selling Price} - \text{Cost Price}$$

Here,

$$\text{Profit} = \text{Rs } 700 - \text{Rs } 400 = \text{Rs } 300$$

Now

$$\text{Profit Rate} = \frac{\text{Profit}}{\text{Cost Price}} \times 100$$

$$\text{or Profit Rate} = \frac{300}{400} \times 100 = 75\%$$

Thus the profit rate is 75% and his profit is Rs 300.

Example 2: A computer software retailers uses a profit rate of 45%. Find the selling price of an electronic computer game that costs the retailer Rs 1250.

Solution:

We know that

$$\text{Selling price} = \text{Cost price} (1 + \text{Profit Rate})$$

Here,

$$\text{Selling price} = ?$$

$$\text{Cost price} = \text{Rs } 1250$$

$$\text{Profit Rate} = 45\% = 0.45$$

$$\text{S.P} = \text{C.P} (1 + \text{Profit Rate})$$

$$\text{So, Selling Price} = \text{Rs } 1250 (1 + 0.45) = \text{Rs } (1250 \times 1.45) = \text{Rs } 1812.50$$

Thus the Selling Price of the computer game is Rs 1812.50

Example 3: A shoes store uses a 40% profit rate on cost price. Find the cost of a pair of shoes that is sold for Rs 2,630.

Solution:

We know that

$$\text{Selling price} = \text{Cost price} (1 + \text{Profit rate})$$

$$\text{Here, Cost price} = \text{CP} = ?$$

$$\text{Selling price} = \text{Rs } 2630$$

$$\text{Profit Rate} = 40\% = 0.4$$

$$\text{S.P} = \text{C.P} (1 + \text{Profit Rate})$$

$$\text{So, } 2630 = \text{CP} (1 + 0.4)$$

$$\text{or } \text{C.P} = \text{Rs } \frac{2630}{1.4} = \text{Rs } 1,878.57$$

Hence the cost of a pair of shoes is Rs 1,879.

Remember: The profit is also calculated as a percentage of the original cost price.

$$\text{Profit Rate} = \frac{\text{Profit}}{\text{Cost Price}} \times 100$$

Profit is used to determine the price in which the goods should be sold.

Thus there are in fact two different ways to analyze the cost and profit of a product or service in small business.

Example 4: A Tile company has Rs 1,500,000 of sales in the previous month. As well as sale returns (means the sale returned back by the consumer due to some reasons) of Rs 45,000, a cost of goods sold for Rs 650,000 and administrative expenses of Rs 470,000. Calculate the net profit percentage of the company.

Solution:

$$\text{We know that: Profit Rate} = \frac{\text{Gross Profit}}{\text{Net Sale}} \times 100$$

Here, Net Sale = Rs 1,500,000 (Sales) – Rs 45,000 (Sale returned)
= Rs 1,455,000

Gross Profit = Rs 1,455,000 (Net Sale) – [Rs 650,000 (Cost of goods)
– Rs 470,000 (Administrative expenses)]
= Rs 335,000

Formula: Profit Rate = $\frac{\text{Gross Profit}}{\text{Net Sale}} \times 100$

So, Profit Rate = $\frac{335,000}{1,455,000} \times 100 = \frac{335,000 \times 100}{1,455,000} = \frac{6700}{291} = 23\%$

The profit percentage of Tile Company is 23%.

EXERCISE 7.3

1. A retailer pays to its whole-saler Rs,1860 for a Helmet and then he sells it to consumer for Rs 2,650. What is the markup rate and his profit.
2. A Shopkeeper sells a microwave oven for Rs 6700. Find the markup rate and his profit if the cost price of microwave oven is Rs 6250.
3. A retailer uses markup rate of 65% on the computer accessories. Find the selling price if the cost of a certain accessory is Rs 1520.
4. The cost price of a certain novel is Rs 1,100. Find selling price of the novel at 66% markup rate.
5. A Mobile Phone company uses a markup rate of 55% on the cost of each mobile phones that cost Rs 2,260. Find the selling price of the mobile phone.
6. A mobile phone company uses a 45% markup rate. Find the cost of a unit that sells for Rs 4785.
7. The markup rate on the selling price of a calculator that is Rs 1,230 is 50%. Find the cost price of the calculator.
8. A stationer uses a markup of 60% on each item that he sells. Rs 144. Find the cost price of the item.

9. A home-based small business brings in Rs 25000 a month in sales of baby clothing and accessories. The cost of goods sold including overhead add up to Rs 10550. Calculate the net profit percentage.
10. A certain business has Rs 2,500,000 of sales in one month. Sales returns of Rs 15,000. The goods sold were of cost 505,000 and overhead expenses of Rs 580,000. Calculate the profit rate percent.
11. A medicine company has 1,850,000 of sales in the previous month. The sale returns of Rs 85,000. The medicines sold were of cost 850,000. The administrative expenses are of Rs 550,000. Calculate the net profit percentage of the company.

7.2.3 (ii) Solve real life problems involving markup.

Suppose '**P**' is the principal amount, '**T**' is the time period and '**R**' is the markup rate. Then the amount of **markup** will be:

$$\text{Markup} = \frac{R \times P \times T}{100}$$

In the same way we can find:

$$\text{Principal Amount} = \frac{\text{Mark up} \times 100}{T \times R}, \text{ Rate of markup} = \frac{\text{Mark up} \times 100}{P \times T}$$

$$\text{and Time period} = \frac{\text{Mark up} \times 100}{P \times R}$$

Consider the following examples:

Example 1. A person borrowed a loan of Rs 280,000 from a bank at the markup rate of 12% per annum. Find the value of markup, if the time period is 2 years.

Solution:

Principal amount = Rs 280,000 Markup rate = 12% annum

Time period (T) = 2 years Value of Markup = ?

Formula:

$$\text{Value of the markup} = \frac{R \times P \times T}{100} = \frac{12 \times 280,000 \times 2}{100} = \text{Rs } 67,200$$

Thus the value of the markup is Rs 67,200.

Example 2. Aslam borrowed a loan of Rs 450,000 from a retail organiser for 292 days at the rate of 10% per annum. Find the value of markup.

Solution:

Principal (P) = Rs 450,000, Markup rate (R) = 10% per annum, Markup = ?

Time period = 292 days = $\frac{292}{365}$ year = $\frac{4}{5}$ year

By using the formula: Markup = $\frac{R \times P \times T}{100}$

$$= \frac{10 \times 450,000 \times \frac{4}{5}}{100} = \text{Rs } \frac{10^2 \times 450,000 \times 4}{100 \times 5} = \text{Rs } 36,000$$

Thus the value of markup is Rs 36,000.

Example 3. On a bank loan, calculate the markup on a principal amount of Rs 87,600 for 20 months at the rate of 12.5% per annum.

Solution:

Principal amount (P) = Rs 87600 Markup rate (R) = 12.5%

Time period (T) = 20 months = $\frac{20}{12}$ year = $\frac{5}{3}$ years

Markup = ?

By using the formula: Markup = $\frac{R \times P \times T}{100}$

$$\text{Markup} = \frac{12.5 \times \frac{5}{3} \times P}{100} = \frac{12.5 \times 5 \times \frac{87600}{3}}{100 \times 3} = 18250$$

Hence the markup is Rs 18,250

EXERCISE 7.4

1. Qasim bought a car for Rs 450,800 from a bank loan on instalments at the markup rate of 12% per annum. Find the sale price of the car, if the time period is 3 years.
2. Najma bought a golden ornament set for Rs 1,75,500, on installments at the markup rate of 10% per annum. Find her sale price of the set if the time period is 9 months.
3. Shazia sold his mobile phone for Rs 16,900 and got 15% markup rate per annum. Find her markup where the period is 219 days.
4. If 15% markup on a colour T.v set is 8,500 for 1½ years. Find the cost price of the colour T.v set.

5. Find the missing quantities by using formula:

S. No.	Markup	Principal	Time period	Markup Rate
(i)	Rs _____	Rs 800,000	1 year	12%
(ii)	Rs 10000	Rs 500,000	_____	10%
(iii)	Rs 6,600	_____	2 years	11%
(iv)	Rs _____	Rs 200,000	1½ years	8%
(v)	Rs 800	Rs 4,000	2½ years	
(vi)	Rs 45,000	_____	73 days	10%

7.3 ZAKAT AND USHR

Zakat and Ushr Ordinance 1980

In Pakistan, Zakat Councils are responsible for collecting and distributing Zakat and Ushr. The Ministry of Religious Affairs oversees the Zakat Councils. The collection and distribution of Zakat and Ushr system was enforced in 1980.

7.3.1 Define zakat and ushr

(i) Zakat

Zakat is one of the five pillars of Islam. It has been mentioned along with Salat in Quran.

Zakat is obligatory on every Sahib-e-Nisab muslim who possesses wealth in excess of Nisab that is 87.48 grams or 7.50 tola Gold; or 612.36 grams or 52.50 tola silver, excluding his needs for a complete lunar year.

The zakat is paid after a complete lunar year at the rate of 2.5%.

$$\text{Zakat} = \text{Annual saving} \times 2.5\%.$$

(ii) Ushr

Ushr is paid on compulsory basis at the rate of 5% (or 10%) of the produce by every land owner, grantee, allottee, lessee, lease-holder or land holder.

An individual farmer is exempted from ushr if he is a 'mustahiq' or his produce is less than 'Five Wasaqs or 948 kg (approximately) of wheat or its equivalent in value'.

If the farm or garden is watered naturally by rain or spring water or river or stream water, tenth part or 10% is obligatory, and if farm or garden is irrigated by artificial means like well, tube well, canals etc. twentieth part or 5% is obligatory.

(iii) Distribution of Zakat and Ushr

Zakat is paid:

- to the poor and needy.
- to those who are in bondage and debt and for wayfarer.
- to those who are employed to administer the funds.

7.3.2 Solved problems related to zakat and ushr.

Example 1: After a whole year a business man has saved Rs 80,000 in cash. He has a cash in Bank of Rs 600,000. Find the Zakat due on him.

Solution:

(i) Saving in cash = Rs 80,000

(ii) Cash in bank = Rs 600,000

Total saving of the businessman = 80,000 + 600,000

= Rs 680,000

Amount of Zakat due on him = Rs 680,000 × 2.5%

= $680,000 \times \frac{2.5}{100}$ = Rs 17,000

the Zakat due on the businessman is Rs 17,000.

Example 2: A household woman has gold ornaments of value of Rs 500,000. She also saved an amount of Rs 25,000 from daily life expenses. Find the Zakat due upon her.

Solution:

Total amount = Rs 500,000 + Rs 25,000 = Rs 525,000

So, Zakat = Rs 525,000 × 2.5 %

= $Rs\ 525,000 \times \frac{2.5}{100}$ = Rs 13,125

Thus, she has to pay Rs 13,125 as Zakat.

Example 3: Farah paid an amount of Rs 9,000 as Zakat. Find the saving on which she paid the Zakat.

Solution:

We know that: Amount of Zakat = Annual saving $\times$ 2.5%

$$9,000 = \text{Annual saving} \times 0.025$$

$$\text{Annual saving} = \frac{9,000}{0.025} = \text{Rs } 360,000$$

Thus she had an annual saving of Rs 360,000.

Example 4: A landlord produces 3,000 kg of wheat and vegetables of worth Rs 25,000. Find the amount of Ushr due upon him at the rate of 5% and the rate of wheat is Rs 30 per kg.

Solution:

So,

$$\text{Ushr} = \text{Total value of Produce} \times 5\%$$

$$\text{Value of 3,000 kg wheat at Rs 30 per kg}$$

$$= \text{Rs } (3,000 \times 30) = \text{Rs } 90,000$$

$$\text{Total value of wheat and vegetable} = \text{Rs } 90,000 + \text{Rs } 25,000$$

$$= \text{Rs } 115,000$$

$$\text{Amount of Ushr} = \text{Rs } 115,000 \times 5\%$$

$$= \text{Rs } 115,000 \times \frac{5}{100} = 115,000 \times 0.05$$

$$= \text{Rs } 5,750$$

The landlord has to pay Rs 5,750 as Ushr.

EXERCISE 7.4

1. A person holds an amount of Rs 500,000 for the whole year. Is he sahib-e-nisab? If yes, find the amount of Zakat he has to pay.
2. A salaried person possess Rs 200,000 and 3 tolas of pure gold. Find the Zakat to be paid by him. Rate of gold is Rs 47,400 per tola.
3. A lady doctor possesses ornaments of worth Rs 500,000 and saved Rs 150,000 in the whole year. Find the amount of Zakat due upon her.

4. A business man paid Rs 20,500 as Zakat. Find his saving for the whole year.
5. A farmer pays Rs 10,000 as Ushr. Find the value of the crop he produced. Rate of Ushr is 5%.
6. A farmer produces 15,000 kg of sugarcane. Find his amount of Ushr at the rate of 5%. The rate of sugarcane is Rs 5.25 per kg.
7. Jumman exports oranges of his garden for Rs 5,000,000. Find the amount of Ushr at the rate of 5%.
8. Kamil produces 1,500 kg of wheat on one part of his land and 1,200 kg of rice on the other part of his land. Find the amount of Ushr at the rate of 5%. The rate of wheat is Rs 30 per kg and rate of rice is Rs 65 per kg.

REVIEW EXERCISE 7

1. The cost of a mobile phone is Rs 8,500. The seller charges 17% GST on the cost. Find sales tax levied.
2. An imported LCD TV is sold for Rs 55,800 inclusive of GST. Find the cost of LCD excluding tax. GST rate is 17%.
3. An electric company charges Rs 16,750 as electric charges from consumer. Find the amount of GST at the rate of 17%, paid by the consumer.
4. A person paid Rs 16,989 as GST at the rate of 17% on the purchase of room-set. Find the cost of room set exclusive of GST.
5. A business man gets loan of Rs 1,700,000 from a bank at the markup rate of 12.5% per annum. Calculate the markup for a period of $1\frac{1}{2}$ years.
6. The markup paid to a bank by a lady doctor is Rs 2850. The period is 6 months and the markup rate is 12%. Find principal amount
7. A senior citizen deposits an amount of Rs 500,000 in a bank. Find the his markup when the markup rate is 10% and the time period is $2\frac{1}{2}$ years.

8. A car dealer uses 35% markup rate on costs. Find the cost of a car that sells for Rs 1,051,000.
9. After a whole year a business man saved Rs 95,000 and has a stock of Rs 500,000. Find the zakat due on him.
10. A lady teacher has gold ornaments of value 400,000 rupees. Her annual is an amount of Rs 45,000. Find the Zakat due upon her.
11. A gentleman paid an amount of Rs 12,500 as Zakat. Find his amount on which he paid the Zakat.
12. A farmer produces 5,000 kg of wheat and vegetables of worth Rs 48,000. Find the amount of Ushr due upon him at the rate of 5%. The rate of wheat is Rs 35 per kg.
13. A farmer pays Rs 2,000 as Ushr. Find the value of the crop he produced. Rate of Ushr is 5%.

SUMMARY

- ➔ TAX is a fee charged by a government on a product, income, or
- ➔ THE PURPOSE of taxation is to finance government expenditures.
- ➔ PROPERTY TAX is a fee charged by a government on a person's
- ➔ GENERAL PROPERTY TAX = 2% of taxable income
- ➔ GST = Item Cost $\times$ Rate of GST.
- ➔ Rate of GST in Pakistan is 17%.
- ➔ TOTAL ITEM COST = General Sales Tax (GST) + Item Cost.
- ➔ ITEM COST = $\frac{\text{Total Cost}}{1.17}$
- ➔ PROFIT = Selling Price – Cost Price.
- ➔ Profit Rate = $\frac{\text{Profit}}{\text{Cost Price}} \times 100$

- ➔ Markup Rate = $\frac{P \times R \times T}{100}$ where P means principal amount, T means time period. R means markup rate per annum.
- ➔ SELLING PRICE = Cost Price (1 + Markup Rate)
- ➔ PROFIT RATE = $\frac{\text{Gross Profit}}{\text{Net Sale}} \times 100$
- ➔ ZAKAT is obligatory on every sahib-e-nisab muslim. It is an Islamic tax which is paid on the wealth which remains with a person for a complete years.
- ➔ NISAB: Gold 87.48 grams or 7.5 tola
or Silver 612.36 grams or 52.5 tola
- ➔ ZAKAT RATE is 2.5%
- ➔ USHR is paid by the farmer or land owner.
- ➔ USHR NISAB is value of 948 kg wheat or 5 wasaqs.
- ➔ USHR RATE is 5% or 10% of the land produce.

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Think Tank: The Most Wonderful Knowledge.

Can you imagine that you can get the same result even if you change the style of operation?

It is unimaginable but still you will observe that

$$1\frac{1}{2} \times 3 = 1\frac{1}{2} + 3; \quad 3\frac{1}{5} + \frac{4}{5} = 3\frac{1}{5} \div \frac{4}{5}$$

$$7\frac{1}{5} \div 6 = 7\frac{1}{5} - 6; \quad 5 - \frac{5}{6} = \frac{4}{5} 5 \times \frac{5}{6}$$

Be careful! Don't make it a rule every time.